



ABOUL DAHAB

LOCAL AI AUDIO TOOLSET

Music Studio

User Manual

Any song in. Lyrics, tabs and karaoke out.

Subtitles | Karaoke & Lyric Videos | Chord Sheets

Guitar Tabs | Full Transcription | AI Music

Version 3.2.0
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Designed & programmed by Amr Aboueldahab
Runs entirely on your own machine - no cloud upload, no subscription.

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1 Introduction

What Music Studio is and what it can do for you

What is Music Studio?

Music Studio is a private, browser-based workstation that turns any song into synchronised lyrics, karaoke videos, chord sheets, guitar tabs and more. You drop in an MP3, MP4 or other audio or video file; the program listens to it with local AI models, and hands back ready-to-use files.

Everything runs on your own computer. Your recordings are **never uploaded to the cloud**, and there is no subscription. The only time the internet is needed is the first run of each AI model, which downloads its model files once.

Feature overview

AI Subtitles (SRT)

Whisper transcribes the lyrics with word-level timing, auto language detection and smart cue splitting. Instrumental sections are labelled automatically.

Karaoke & Lyric Videos

Burns synced lyrics onto your own image or video background (or an automatic backdrop). Karaoke removes the vocals; Lyric Video keeps them.

Lyrics, Chords & MP3

A print-ready lyric sheet with chords above the words and chord diagrams, plus an MP3 copy of the track. Supports Arabic and other right-to-left text.

Guitar Solo Tab

Pitch tracking converts a guitar solo into a printable tab with strings, frets and expressive techniques (bends, vibrato, hammer-ons).

Full Transcription

One click produces guitar TAB, chord charts, MIDI files, piano sheet music and a fingerstyle (Travis picking) tab, plus a text report.

AI Music Generation

Write lyrics, choose a style, singer and instruments, and generate an original song with ACE-Step (with vocals) or MusicGen (instrumental).

Built for teams and shared computers

- **Web dashboard** - works in any modern browser, also from other devices on your local network.
- **Multi-user accounts** with an admin panel and role management; each user's files are kept separate.
- **Live progress** with status text, a timer and a progress bar for every job.
- **Persistent jobs** - reload the page or restart the server and your job is picked back up.
- **One-click Browse buttons** open your output folders in Explorer / Finder.

Who is it for?

Content creators, worship and choir teams, karaoke hosts, guitar instructors, video editors and songwriters - anyone who needs accurate lyrics, timing, chords or tabs from a recording.

2

Getting Started

Requirements, starting the server, and reaching the app

What you need

Music Studio is packaged as a Python web application (or a Windows .exe built from it). Some components are required for the core features; others are optional and only unlock specific tools.

Component	Needed for	Status
FFmpeg (on the system PATH)	Reading video files, audio conversion, vocal removal, rendering videos, MP3 export	Required
Whisper (openai-whisper)	Transcribing lyrics - the heart of Generate Subtitles	Required
Demucs	Vocal isolation, karaoke instrumental, guitar stem extraction	Required for karaoke and guitar isolation
stable-ts	More accurate word timing on sung vocals (better karaoke sync)	Strongly recommended
librosa, numpy, reportlab	Chord detection, guitar tabs and all PDF sheets	Required for chords / tabs
madmom	The <i>madmom</i> and <i>hybrid</i> chord methods	Optional
MUSIC.py, music21, soundfile	Full Transcription	Required for that feature only
MuseScore	The piano sheet PDF inside Full Transcription	Optional (sheet skipped without it)
ACE-Step runtime	AI Music Generation with vocals (bundled in the packaged app)	Optional
transformers, torch, scipy	AI Music Generation with the MusicGen backend (instrumental)	Optional
waitress	Faster production web server (falls back to Flask's built-in server without it)	Recommended

Starting the program

1

Launch it

Run `app.py` with Python, or double-click the packaged .exe. A console window prints the addresses the server is listening on.

2

Wait for the browser

Your default browser opens automatically once the server is ready. If it does not, open `http://127.0.0.1:8000` yourself.

3

Open the studio

Click **Launch Music Studio** on the welcome page to reach the sign-in screen.

Server	Address	When
Production (waitress)	http://127.0.0.1:8000	Default when waitress is installed
Flask development server	http://127.0.0.1:5000	Automatic fallback if waitress is missing
From another device on your network	http://<this-computer's-LAN-IP>:8000	The console prints the exact LAN address

NOTE AI music server starts by itself

When the program starts it also launches the bundled ACE-Step server quietly in the background (used only by AI Music Generation). The first start can take 30-120 seconds while models load. To turn this off, set the environment variable `ACE_AUTOSTART=0` before starting.

First-time account

On first run the program creates an administrator account so you can sign in:

Username	Password	Role
admin	change-me-now	admin

WARNING Change the default password immediately

Anyone who knows the default password can sign in as administrator. After your first login, create your own admin account (see **Administration**) and remove or re-password the default one.

3

Signing In

Accounts, sessions and logging out

The sign-in screen

Access to Music Studio is by invitation: accounts are created by an administrator. Enter your username and password and press **Sign In**. On success you are redirected to the dashboard; on failure a red message appears (*Invalid username or password*).

- **Stay signed in:** your session lasts **7 days**, even if you close and reopen the browser.
- **Already signed in?** Opening the site takes you straight to the dashboard.
- **Logout:** use the **Logout** button at the top-right of any page (next to your username).
- **No account?** Ask your administrator to add one.

What each role can do

	user	admin
Use every dashboard tool (subtitles, videos, tabs, transcription, AI music)	Yes	Yes
See and download their own files	Yes	Yes
Follow the status of other users' jobs	No	Yes
Open the User Management page and add / delete accounts	No	Yes

4 The Dashboard at a Glance

A guided tour of every panel and button

After signing in you land on the dashboard. It is one page: settings and buttons on the left, a live subtitle preview on the right. The numbers in the diagram match the descriptions below.

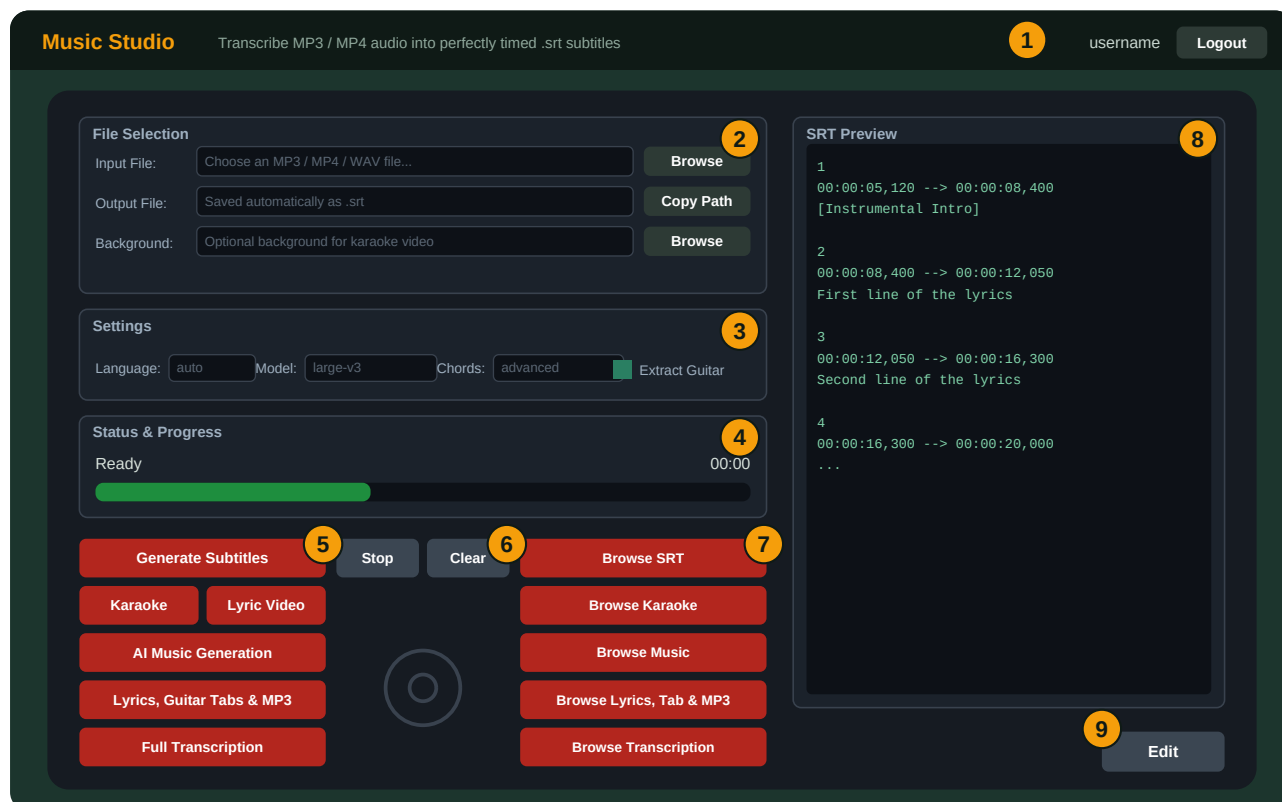


Figure 1 - The dashboard. Button colours change as jobs run (see the colour legend below).

#	Area	What it does
1	Header bar	Shows the app name, your username and the Logout button.
2	File Selection	Input File: pick the song. Output File: shows the .srt name (Copy Path copies it). Background: optional image or video for karaoke / lyric videos.
3	Settings	Language , Whisper Model , Chords method, and the Extract Guitar checkbox.
4	Status & Progress	Live status message, elapsed timer and a progress bar for the running job.
5	Action buttons	Generate Subtitles, Karaoke, Lyric Video, AI Music Generation, Lyrics / Guitar Tabs & MP3, Full Transcription.
6	Stop / Clear	Stop cancels the running job. Clear resets fields, status and button colours.
7	Browse buttons	Open the folders where each kind of result is saved.
8	SRT Preview	Shows the generated subtitles as soon as they are ready.
9	Edit	Opens the subtitles for editing and saving.

Button colour legend

The six job buttons tell you their state at a glance:

Red	Ready - idle and available.	Amber	Processing - its job is running.	Green	Done - finished (until used again or Clear is pressed).
------------	------------------------------------	--------------	---	--------------	--

TIP Only one thing at a time is needed

You can leave the page while a job runs. Jobs are remembered for 24 hours, so when you come back (or after a server restart) the dashboard restores your session and the SRT preview.

5

Quick Start

From song to finished files in five steps

Main pipeline

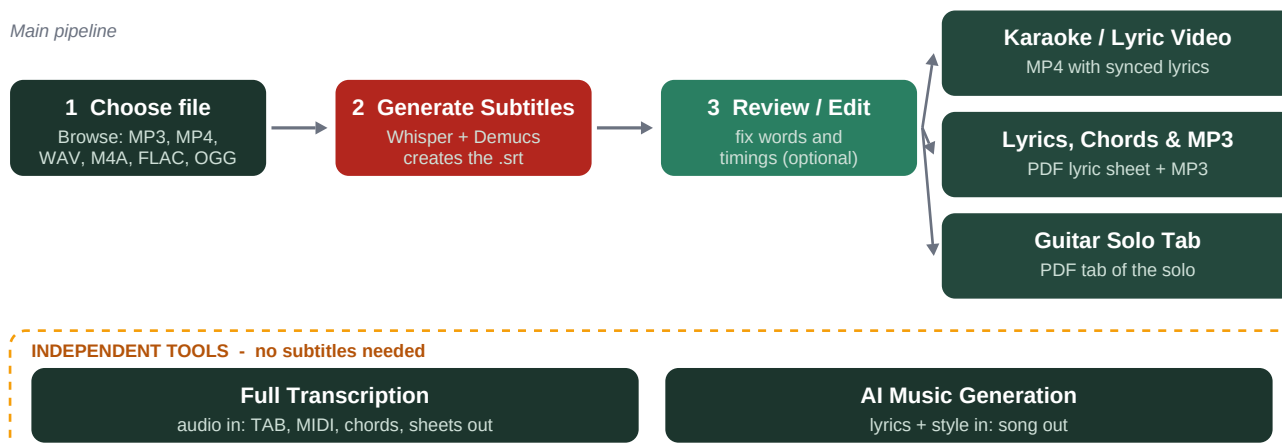


Figure 2 - How the tools fit together. Karaoke, lyric sheets and tabs build on the subtitles from step 2.

- 1 Choose your song**
Under **File Selection**, click **Browse** next to Input File and pick an MP3, MP4 or other audio/video file. It is copied to your private upload folder and the Output File box shows the matching .srt name.
- 2 Set your options**
Pick the song's **Language** (or leave *auto*), keep the **Model** on *large-v3* for best accuracy, and leave **Chords** on *advanced*.
- 3 Click Generate Subtitles**
The button turns amber and the progress bar starts. When it turns green (Status: *Complete!*), your subtitles appear in the **SRT Preview**.
- 4 Check and edit**
Read through the preview. Press **Edit** to correct any word or timing before you build anything from it.
- 5 Create what you need**
Click **Karaoke** or **Lyric Video** for a video, **Lyrics, Guitar Tabs & MP3** for the printable sheets, then use the matching **Browse** button to open the results.

TIP Want tabs and chords only?

Full Transcription works straight from the audio - no subtitles required.

Supported file formats

Purpose	Accepted formats
Input song (Browse in File Selection)	MP3, MP4, WAV, M4A, FLAC, OGG. Video files are converted to audio automatically with FFmpeg.
Background for karaoke / lyric video (optional)	Images: PNG, JPG, JPEG, BMP. Video: MP4, MOV, MKV, WEBM, AVI.
Files you get back	SRT, MP4, PDF, MP3, WAV, MIDI, TXT

6 Feature Guide

Detailed instructions for every tool

Generate Subtitles

NEEDS An input file	PRODUCES A timed .srt subtitle file, shown in SRT Preview	SAVED IN outputs/<you>/SRT/<song>/
-------------------------------	---	--

This is the foundation of the studio. It listens to the song and writes a standard SRT subtitle file, one cue per lyric line, each with a start and end time.

How to use it

- 1 Select the input file**
Browse and choose your song.
- 2 Choose Language and Model**
See **Settings Reference** for guidance. The default (auto / large-v3) works well for most songs.
- 3 Press Generate Subtitles**
Watch the status line. The first run of a model downloads it, which can take a while.

What happens behind the scenes

Progress	Stage
5 - 10 %	Loading the Whisper model and preparing the audio (video files are converted to audio).
15 %	Vocal isolation with Demucs, so instruments do not confuse the transcription. If Demucs or FFmpeg is unavailable, the program automatically continues with the full mix.
25 - 85 %	Transcribing - the slowest step, with word-level timestamps.
85 - 100 %	Formatting the subtitles and saving the .srt.

Words are grouped into cues that break at natural pauses, sentence ends, or when a line gets too long. Long gaps are filled with [Instrumental Intro], [Instrumental] and [Instrumental Outro] markers, and phantom text that Whisper invents over pure music is replaced with [Instrumental].

TIP For the best karaoke sync

Make sure **stable-ts** is installed. Plain Whisper's word timing can drift on sung vocals; stable-ts refines it. If it is missing, the status line tells you.

Editing subtitles

Press **Edit** (below the SRT Preview) to correct the text or timing of any cue, then save. Saving writes the changes back to the same .srt file in your output folder, so every video or sheet you create afterwards uses the corrected version.

SRT anatomy	
1	Cue number
00:00:08,400 --> 00:00:12,050	Start and end time (hours:minutes:seconds,milliseconds)
First line of the lyrics	The text shown during that time. A blank line separates cues.

Karaoke video

NEEDS Generated subtitles; optional background	PRODUCES MP4 karaoke video without vocals	SAVED IN outputs/<you>/Karaoke/<song>/
--	--	--

- 1 Generate subtitles first**
Karaoke needs the SRT. If none exists you will see *SRT not found - generate subtitles first*.
- 2 Optional: choose a background**
Use **Background Browse** to pick an image or video. Leave it empty and the program creates a soft cloud backdrop for you.
- 3 Press Karaoke**
Demucs strips the vocals (about 20 %), then FFmpeg renders the video with the synced lyrics (from about 60 %).
- 4 Open the result**
Use **Browse Karaoke**. The file is named <song>_karaoke.mp4.

Lyric Video

NEEDS Generated subtitles; optional background	PRODUCES MP4 video with the original vocals	SAVED IN outputs/<you>/Karaoke/<song>/
--	--	--

Identical to Karaoke except that the original song is kept as-is - ideal for lyric videos you want to publish. It is saved in the same Karaoke folder with the name <song>_lyric_video.mp4, so the one **Browse Karaoke** button reaches both.

Lyrics, Guitar Tabs & MP3

NEEDS Generated subtitles; chord method; Extract Guitar setting	PRODUCES Lyric-sheet PDF, MP3 copy and a solo-tab PDF	SAVED IN Lyrics/<song>/ and Tabs/<song>/
---	---	--

One button, two exports run back to back as a single job (first half of the progress bar for the lyrics, second half for the tab).

Part 1 - Lyrics sheet + MP3

- Each subtitle line is printed with the **detected chord above it**, followed by a chord-diagram reference section.
- Arabic and other right-to-left lyrics are laid out correctly.
- An MP3 copy of the track is saved next to the PDF.

- Chord accuracy depends on the **Chords** setting (see Settings Reference).

Part 2 - Guitar solo tab

- Pitch tracking finds the notes of the solo and assigns strings and frets automatically.
- Techniques such as bends, vibrato and hammer-ons are marked, and notes are grouped into readable phrases.
- With **Extract Guitar** ticked, Demucs first isolates the guitar so the tab follows the guitar rather than the whole mix.
- If no solo range is given, the whole track is analysed (from 0:00 to the end).

NOTE Chords are an aid, not gospel

Detection recognises major, minor and a few dominant-7th chords. Complex jazz voicings and heavily distorted guitars may be simplified. Always check the sheet against the recording.

Full Transcription

NEEDS Just the audio file - no subtitles needed	PRODUCES Guitar tab, chord chart, MIDI, piano sheet, fingerstyle tab, report	SAVED IN outputs/<you>/Transcription/<song>/
---	--	--

A complete musical write-up of the song in one click. Before it starts, the program checks that everything it needs is present and tells you exactly what is missing (MUSIC.py, required Python packages, or FFmpeg).

Progress	Stage
5 %	Convert the input to WAV.
15 %	Separate instruments with Demucs (if this fails, the full mix is used).
30 %	Detect the tempo (BPM) and length.
45 %	Extract the guitar melody - the longest step, it can take several minutes . The status line shows elapsed time so you know it is alive.
60 %	Detect chords.
70 - 75 %	Write the guitar and chord MIDI files.
85 - 88 %	Generate the guitar TAB PDF and the chord / rhythm PDF.
91 %	Piano sheet music (needs MuseScore; skipped if unavailable).
95 %	Fingerstyle guitar TAB, Travis picking (PDF).
97 %	Write the text report.

When finished you will find these files:

File	Contents
01_GUITAR_SOLO_TAB.pdf	Guitar melody as tablature
02_RHYTHM_CHORDS.pdf	Chord chart with rhythm
03_GUITAR_SOLO.mid	Guitar melody as MIDI

File	Contents
04_RHYTHM_CHORDS.mid	Chords as MIDI
05_PIANO_SHEET.pdf	Piano arrangement (only if MuseScore is installed)
06_FINGERSTYLE_GUITAR_TAB.pdf	Travis-picking fingerstyle arrangement
01_INPUT_AUDIO.wav / 02_INSTRUMENTAL_STEM.wav	The converted input and the separated instrumental stem
TRANSCRIPTION_REPORT.txt	Summary: tempo, notes, chords, duration

NOTE "Complete (skipped: ...)"

If the piano sheet or fingerstyle tab could not be produced, the job still finishes and lists what was skipped. Everything else is delivered.

AI Music Generation

NEEDS Lyrics and a style description	PRODUCES A generated song (WAV) plus its lyrics and settings	SAVED IN outputs/<you>/Songs/<title>_<time>/
--	--	--

Press **AI Music Generation** to open the song-creation window. You describe the song; the program generates it locally.

Field	What to enter
Title	Names the song folder and file. Left empty, it is called <i>Untitled Song</i> .
Lyrics	The words to be sung (required for ACE-Step). Use section tags on their own line: [verse], [chorus], [bridge], [outro].
Style	Genre and mood in plain words, e.g. <i>warm acoustic pop, slow tempo</i> .
Singer	<i>auto</i> , a chosen voice type, or <i>instrumental</i> for no vocals.
Instruments	Pick the instruments you want to hear.
Backend	ACE-Step (sings your lyrics) or MusicGen (instrumental only).

Choosing a backend

	ACE-Step (default)	MusicGen
Vocals	Yes - sings your lyrics	No - instrumental only; lyrics and singer are ignored
Length	Chosen automatically from the amount of lyrics (30 s to 6 min)	About 10 seconds - a quick idea sketch
Speed	Slow on CPU (minutes); up to a 90-minute safety limit	Faster and lighter
First use	Models load on the ACE-Step server (30-120 s)	Downloads a ~1.2 GB model once

- **Language:** the singing language is detected from the script you type (e.g. Arabic, Chinese, Japanese, Korean, Russian, otherwise English).
- **Long lyrics:** if your lyrics need more time than the maximum length, a warning shows in the status line - shorten the lyrics or drop repeated sections.
- **Saved setups:** store your title, style, singer, instruments, backend and lyrics under a name and reload or delete them later. Setups are private to your account.
- **Server banner:** if the ACE-Step server is not running, the dashboard shows a warning with a **Start server** button.

TIP Better vocals

Keep lines short (about 7 words), put a section tag before each block, and avoid one huge paragraph. The program re-wraps overly long lines for you, but your own line breaks always sound best.

Browse buttons

The five buttons on the right of the action grid open the folders holding your results:

Button	Opens	Contains
Browse SRT	SRT	Subtitle files
Browse Karaoke	Karaoke	Karaoke and lyric videos (MP4)
Browse Music	Songs	AI-generated songs
Browse Lyrics, Tab & MP3	Lyrics and Tabs	Lyric sheets, MP3 copies and solo tabs
Browse Transcription	Transcription	Full Transcription results

NOTE On this PC vs. another device

Pressed on the computer that runs Music Studio, a Browse button opens the real Explorer / Finder window. A web page cannot open a file window on *another* device, so from a phone or another PC it shows an in-page list of your files with **download** links instead.

Stop and Clear

- **Stop** is active only while a job is running. It marks the job as cancelled and resets the progress display. Cancelling is best-effort: a step already deep inside an AI model may take a moment to wind down.
- **Clear** resets the input fields, status and button colours so you can start on a new song.

7 Settings Reference

Language, model, chord method and Extract Guitar

Language

Choose **auto** to let Whisper detect the language, or pick one yourself for better accuracy on tricky recordings. Available: auto, en (English), ar (Arabic), es (Spanish), fr (French), de (German), it (Italian), pt (Portuguese), ja (Japanese), zh (Chinese). **Tip:** set it whenever you know it - auto-detection can be fooled by a long instrumental intro.

Whisper model

Larger models are more accurate but slower and need more memory. The default is **large-v3**.

Model	Speed	Accuracy	Best for
tiny	Fastest	Basic	Quick tests, clean speech-like vocals
base	Very fast	Fair	Draft transcripts
small	Fast	Good	Everyday use on modest computers
medium	Moderate	Very good	Balanced quality and time
large / large-v2	Slow	Excellent	Difficult audio, multiple languages
large-v3 (default)	Slowest	Best	Final, publish-quality subtitles

Chord detection method

Used by **Lyrics, Guitar Tabs & MP3**. If a chosen method cannot run, the program quietly falls back to a simpler one so you always get a result.

Method	How it works	Notes
basic	Matches the song's pitch content in each lyric line against standard chord templates.	Fast; good for simple songs.
advanced (default)	Blends two pitch analyses, smooths the result and carries the previous chord across gaps.	Best all-round choice.
madmom	Deep-learning chord recognition.	Needs madmom installed; falls back to basic if unavailable.
hybrid	Combines madmom with the basic method.	Try it when advanced misses chords.

Extract Guitar

Ticked by default. When on, Demucs separates the guitar from the rest of the band before the solo tab is created, giving cleaner notes. Turn it off to analyse the full mix (faster, but less clean when other instruments play at the same time).

8 Output Files & Folders

Where everything is saved

Each user has a private workspace. Your originals go to an **uploads** folder, and everything Music Studio creates goes to an **outputs** folder, neatly organised by tool and then by song name.

```
outputs//
  SRT//          .srt
  Karaoke//      _karaoke.mp4
                _lyric_video.mp4
  Lyrics//       _lyrics.pdf
                .mp3
  Tabs//         _solo_tab.pdf
  Transcription// 01_GUITAR_SOLO_TAB.pdf ... TRANSCRIPTION_REPORT.txt
  Songs/_/       .wav, lyrics.txt, song_request.json
  Presets/       .json

uploads//        your original audio / video files
```

- **Privacy:** users only see their own uploads and outputs. Administrators can follow every user's job status.
- **Downloading:** from another device, use the download links shown by the Browse buttons.
- **Housekeeping:** job records are kept for 24 hours and then removed. Your result files are never deleted automatically.
- **Re-running a song:** processing the same song again replaces the files with the same name in its folder.

9

Administration

Managing users and keeping accounts safe

The web admin page

Administrators can open `/admin` in the address bar (for example `http://127.0.0.1:8000/admin`). It lists all accounts and lets you:

- **Add a user** - type a username and password, choose *user* or *admin*, and press Add.
- **Delete a user** - press Delete beside the account and confirm.

The program protects you from lock-outs: you cannot delete your own account, and you cannot delete the last remaining administrator. Non-admins who try to open the page receive an *Admin access required* message.

The desktop User Manager

`User_Management.py` is a stand-alone window for managing accounts directly, without the web server running. It also lets you **change passwords and rename users**, which the web page does not.

```
python User_Management.py (uses users.json next to the script)
python User_Management.py path/to/users.json
```

Button	Action
Add user	Create an account: username, role and password (entered twice).
Edit user	Rename, change role, or tick <i>Change password</i> . Double-clicking a row does the same.
Delete user	Remove the selected account after confirmation.
Verify password	Check whether a password you type matches the stored one - handy for support.

Every change is saved immediately, and a timestamped backup of the previous file is written first (`users.json.bak.<date>`). Make sure it points at the **same** `users.json` that the server uses.

How accounts are stored and protected

- Passwords are never stored as plain text - only salted **scrypt** hashes.
- The whole `users.json` file is **encrypted at rest**. Opening it in a text editor shows unreadable data.
- The encryption key lives beside it in `users.json.key`. **Keep both files together and back up both.** Without the matching key the account list cannot be recovered.
- An older, plain-text `users.json` is upgraded to the encrypted format automatically the next time it is saved.

Security checklist

IMPORTANT Before sharing the studio with others

1. Change the default **admin** password.
2. Replace the placeholder `app.secret_key` in `app.py` with a long random value.
3. Remember the server listens on your whole network, so only run it on networks you trust or behind a firewall.
4. Encryption protects against casual access; on a shared PC also restrict who can read the program folder.

Privacy and network use

- **Your audio stays on your computer.** Transcription, separation, chord and tab analysis all run locally.
- **Internet is needed only** the first time each AI model is used (its files are downloaded once), or if you deliberately point AI Music Generation at a remote ACE-Step server.
- **Network access:** the server accepts connections from other devices on your network, so anyone who can reach it still needs a valid account to sign in.

10 Tips for Best Results

Small habits that make a big difference

Start with clean audio

Studio recordings and full-quality MP3/WAV give the best lyrics and chords. Heavily compressed or noisy recordings work, but need more editing.

Always review the SRT

Fix names, slang and repeated lines *before* making videos or lyric sheets - all later files reuse it.

Pick the language

Selecting the language manually beats auto-detect on songs with a long instrumental intro.

Match model to time

Try *small* for a quick draft, then run *large-v3* for the final version.

Use a matching background

Use a 16:9 image or video for videos that look right on YouTube and TVs.

Be patient with long jobs

Full Transcription's melody step and AI music on a CPU can take minutes. The timer and status text show it is still working.

Recommended workflow for a worship or karaoke night

- 1 Generate Subtitles with the right language and **large-v3**.
- 2 Edit the SRT to perfect the lyrics.
- 3 Create a **Karaoke** video with a branded background for the singers.
- 4 Create **Lyrics, Guitar Tabs & MP3** and print the chord sheet for the band.

11 Troubleshooting

Messages you may see and how to fix them

What you see	Likely cause	What to do
Invalid username or password	Wrong details, or no account yet.	Check spelling/caps; ask an admin to verify or reset your password.
Admin access required	Your account is a normal user.	Ask an administrator, or use an admin account.
SRT not found - generate subtitles first	Karaoke, Lyric Video or Lyrics needs subtitles.	Run Generate Subtitles for the current song first.
ffmpeg is required... / not found on PATH	FFmpeg is not installed or not on PATH.	Install FFmpeg and restart Music Studio.
Status stays on "Loading model" for long	First run is downloading the Whisper model.	Wait; it only happens once per model. Needs internet that first time.
Karaoke error: Demucs did not produce an instrumental	Demucs is missing or failed.	Install/repair Demucs, then retry. Lyric Video does not need it.
MUSIC.py not found next to app.py	Full Transcription script is missing.	Place MUSIC.py in the program folder.
Full transcription requires these packages...	A Python package is missing.	Install the packages listed in the message.
Complete (skipped: piano sheet)	MuseScore not installed.	Install MuseScore if you want the piano sheet; other files are fine.
Stuck on "Extracting guitar melody..."	Normal - it is a long step.	Watch the elapsed timer; wait a few minutes.
Warning banner: ACE-Step not running	AI music server has not started.	Click Start server and wait 30-120 s; check ace_server.log.
HTTP 401 from ACE-Step	API key mismatch between app and server.	Make the key the same on both (ACE_STEP_API_KEY).
MusicGen isn't installed	Optional package missing.	Install transformers, torch and scipy, or use ACE-Step.
Karaoke words out of sync	Plain Whisper timing on sung vocals.	Install stable-ts, regenerate the SRT, or fix timings with Edit.
Browse shows a list, not a folder	You are on another device.	Use the download links in the list.
Cannot decrypt users.json	Key file does not match the data.	Restore the original users.json.key from backup.
Cannot delete a user	It is your own account or the last admin.	Sign in with a different admin, or create another admin first.

TIP When a job fails

The status line shows **Error:** followed by the reason, and the progress bar resets. Read the message, apply the fix above, and press the button again. The console window running the server prints even more detail.

12 Appendix

Quick reference and glossary

Quick reference: which button for what

Button	Needs subtitles?	Result
Generate Subtitles	No (creates them)	SRT file
Karaoke	Yes	MP4, vocals removed
Lyric Video	Yes	MP4, vocals kept
AI Music Generation	No	Generated song (WAV)
Lyrics, Guitar Tabs & MP3	Yes	Lyric-sheet PDF, MP3, solo-tab PDF
Full Transcription	No	Tab, chords, MIDI, piano, fingerstyle, report

Glossary

Term	Meaning
SRT	A simple text format for subtitles: numbered cues, each with start and end times.
Whisper	The speech-recognition AI that turns singing and speech into text.
Demucs	AI that separates a song into stems: vocals, drums, bass, guitar and other.
Stem	One isolated part of a mix, such as vocals only or everything except vocals.
Karaoke track	The song with the lead vocals removed.
TAB (tablature)	Guitar notation showing strings and frets rather than musical staves.
MIDI	A small file describing notes, not sound - open it in any music software.
Travis picking	A fingerstyle pattern where the thumb keeps a steady bass while fingers play the melody.
ACE-Step / MusicGen	The two AI models used to generate music.
FFmpeg	The free tool that converts audio and video and renders the final MP4.

Music Studio v3.2.0

Designed & programmed by Amr Aboueldahab. Runs locally via Whisper, Demucs, FFmpeg, ACE-Step and MuseScore.